

# An Nguyen

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## EDUCATION

### University of Oklahoma

M.S. in Computer Science | GPA: 4.00

B.S. in Computer Science & B.A. in Mathematics | GPA: 3.77

Norman, OK

**Expected:** Fall 2026

**Graduated:** Fall 2025

## SKILLS

**Programming Languages:** Java, Python, PHP, C#, C++, C, JavaScript, TypeScript, R

**Web & Backend:** React, Next.js, Node.js, Express, .NET, REST APIs, Java Swing, Chrome Extensions (MV3), HTML, CSS

**Data, Cloud & Tools:** PostgreSQL, MySQL, MongoDB, Prisma, GCP, AWS, Docker, Git, Bash, AI/ML, CI/CD, Claude Code, Cursor

## EXPERIENCE

### Paycom

Oklahoma City, OK

Software Development Intern (**PHP, SQL**)

May 2025 – August 2025

- Standardized structured logging across **12+** microservices (request IDs, user/action fields) for end-to-end tracing.
- Reduced noisy error log volume by **38%** with MySQL-backed toggle/TTL system for scoped, insightful logging.
- Built admin UI to toggle Monolog processors for targeted troubleshooting without redeployments, saving debug time.

### Paycom

Oklahoma City, OK

Software Development Intern (**C#, .NET**)

May 2024 – August 2024

- Built a background check vendor emulator logging **100%** of request/response traffic to MySQL for deterministic replay.
- Shipped a React/TypeScript investigation console (timeline, filters, payload/status), cutting PM/QA triage time by **~40%**.
- Achieved **~80% coverage** of third-party screening API calls, allowing teams to reproduce vendor behavior.

### U.S. Department of Defense – Tinker Air Force Base

Oklahoma City, OK

Software Engineering Intern (**Java**)

May 2023 – November 2023

- Built a Java Swing maintenance app for **15+** B-2 engineering teams, reducing workflow completion time.
- Implemented Java form validations, cutting invalid submissions by **~35%** and reducing manual review overhead.
- Grew test coverage from **~45%** to **~75%** through automated tests, code reviews, and CI pipeline improvements.

### OU – Gallogly College of Engineering – Undergraduate Research

Norman, OK

Undergraduate Researcher

August 2023 – January 2024

- Conducted a systematic literature review (SLR) on AI/ML in racing games; presented findings at the ASEE conference.
- Synthesized **20+** peer-reviewed studies via structured multi-database search with defined inclusion/exclusion criteria.
- Worked with a faculty mentor to validate methodology and produce publication-ready findings.

## PROJECTS

### F1 Race Strategy Simulator | **Python, FastF1, pandas, numpy, Arcade, Git**

Present

- Extended an open-source F1 simulator to model custom pit strategies (timing, tires, undercuts) against real race data.
- Visualized strategy outcomes and lap deltas through an interactive race timeline.
- Leveraged Claude Code and Cursor for rapid prototyping and iteration.

### LeetCode Recommendation Engine | **Python, React, TypeScript, Node.js, PostgreSQL, Prisma, AWS, Git**

2025

- Designed and trained a Python recommendation ranking system suggesting the next problem by topic and difficulty.
- Shipped a Chrome extension and dashboard tracking LeetCode progress and notes across 150 problems.
- Accelerated development with Cursor, using it for code generation and refactoring across the full stack.

### Business Finance Dashboard | **React, TypeScript, Node.js/Express, MongoDB, Git**

2025

- Developed a financial analytics dashboard using the MERN stack to visualize complex datasets and trendlines.
- Implemented a simple regression model to predict yearly revenue, designed for integration with real business data.
- Built reusable React/TypeScript widgets (KPI cards, time-series, bar/scatter) for fast analytical insight.

## ACTIVITIES AND LEADERSHIP

### Vietnamese Eucharistic Youth Movement

Oklahoma City, OK

Youth Leader, Lion Dancer

Aug 2014 – Present

- Cultivating faith, leadership, and responsibility among **+400** children through religious and cultural education.
- Featured lion dancer at an OKC Thunder NBA game; fundraise annually through Lunar New Year performances.
- Collaborate with fellow youth leaders to plan seasonal camps, weekly lessons, and team-building activities for kids.